

GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.3

John Chitoti

**Lecturer, Pharmacology
School of Medicine
Copperbelt University**

LS 1.2.1- MEMBRANE TRANSPORT AND ABSORPTION

OBJECTIVES

Students should be able to:

- Define absorption processes involved biological membranes across the human body.

MEMBRANE TRANSPORT AND DRUG ABSORPTION

INTRODUCTION:

- Drug absorption is the movement of a drug from its site of administration onto the blood stream
- In many cases, a drug must be transported across one or more biologic membranes to reach the blood stream
- Drugs are transported across biological membranes by the following mechanisms: passive diffusion, filtration, facilitated diffusion, active transport and endocytosis

MEMBRANE TRANSPORT AND DRUG ABSORPTION

LEARNING OBJECTIVES:

At the end of the lecture, students shall be able to describe:

- 1. Drug transport processes across cell membranes**
- 2. Physicochemical determinants; membrane characteristics, drug properties**
- 3. Mechanism of drug transport; Passive diffusion, carrier-mediated transport, bulk flow and filtration**

MECHANISMS OF TRANSPORT OF DRUGS ACROSS BIOLOGICAL MEMEBRANES

Passive transfer

- Passive diffusion
- Filtration

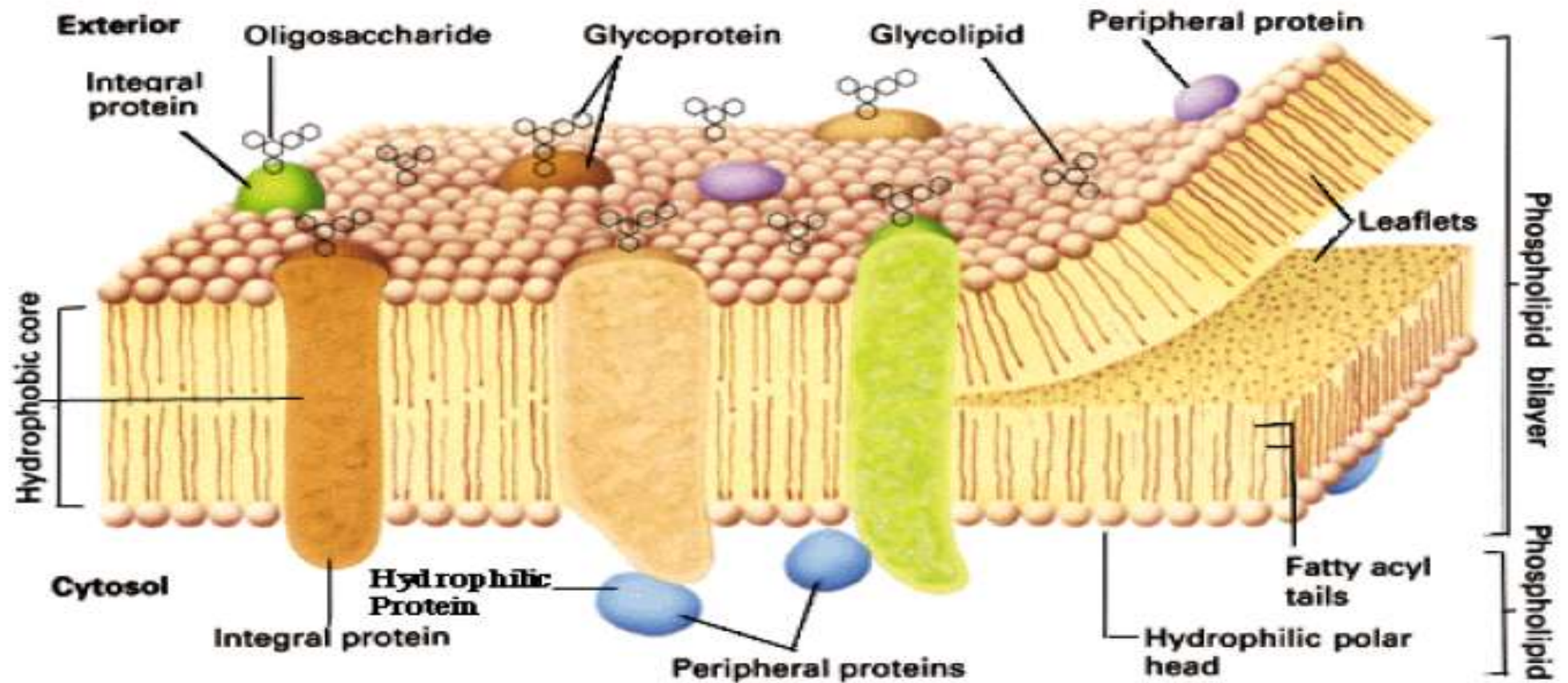
Carrier mediated transport

- Facilitated diffusion
- Active transport

Endocytosis

Membrane characteristics

- Bilayer of amphipathic lipids → **barriers**
- Embedded proteins

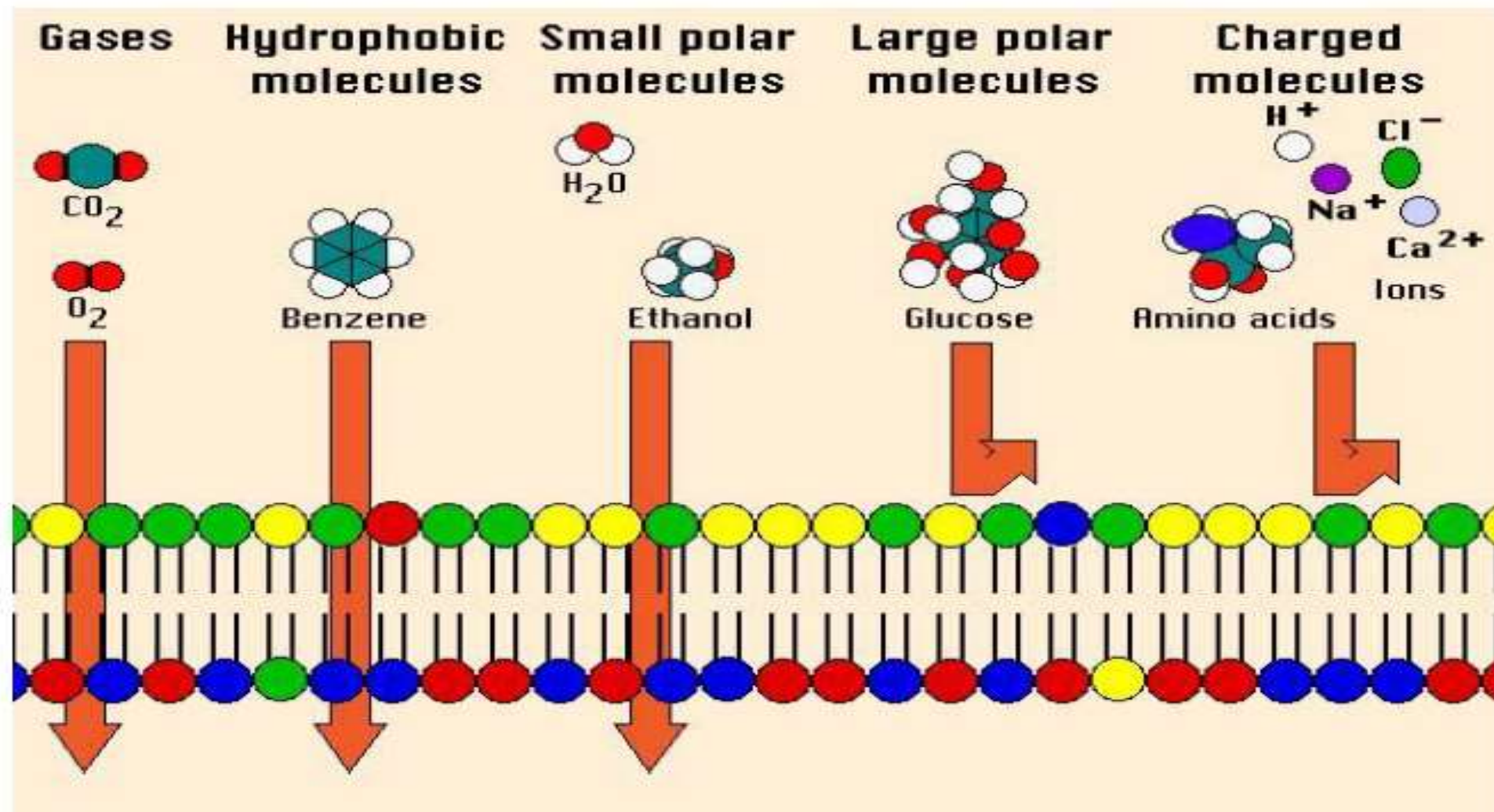


Drug properties

- **Molecular weight, shape and size**
- **Water solubility** (basic property)
- **Lipid solubility**
- **Ionization**

Molecular weight, shape and size

- Small molecules – more chance of crossing membrane

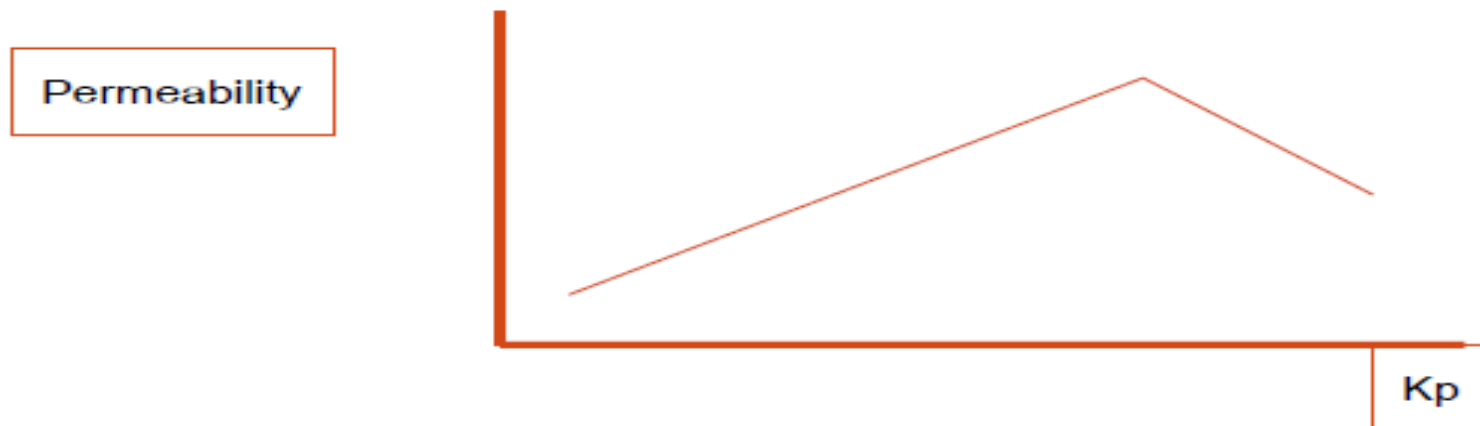


Selective permeability

Lipid Solubility

Movement directly through the lipid bilayer requires that the substance dissolve into the lipid bilayer

Partition coefficient (Kp) = solubility in lipid / solubility in water



Increase lipid solubility causes increased partition coefficient
Increase in partition coefficient causes increased permeability.

Ionization

General rules

- A drug usually exists in 2 forms – unionized and ionized forms
- Unionized drug can passively diffuse across membrane
 - The ratio of unionized drug will indicate direction of passive diffusion
- Factors affecting ionization are
 - pH of the medium
 - pKa (acid dissociation constant) of the drug
- Acidic drug tend to ionize in more basic medium
 - $\text{pH} - \text{pKa} = \log (\text{ionized} / \text{nonionized})$
- Basic drug tend to ionize in more acidic medium
 - $\text{pH} - \text{pKa} = \log (\text{nonionized} / \text{ionized})$

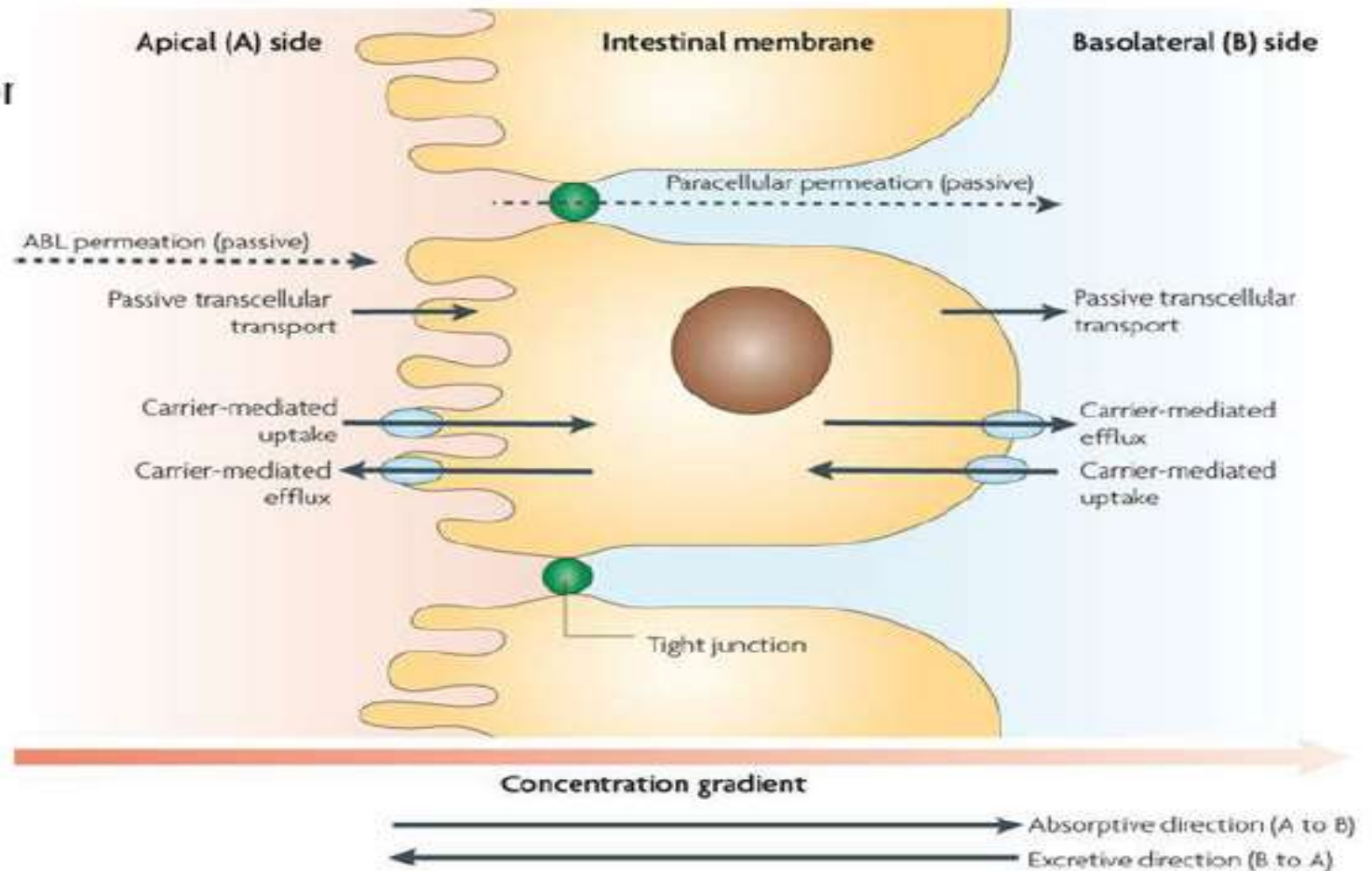
Mechanisms of Drug Transport

Transcellular transport

- Passive diffusion
- Carrier-mediated transport
 - Facilitated diffusion
 - Active transport
- Ion-pair transport
- Endocytosis or Pinocytosis

Paracellular transport

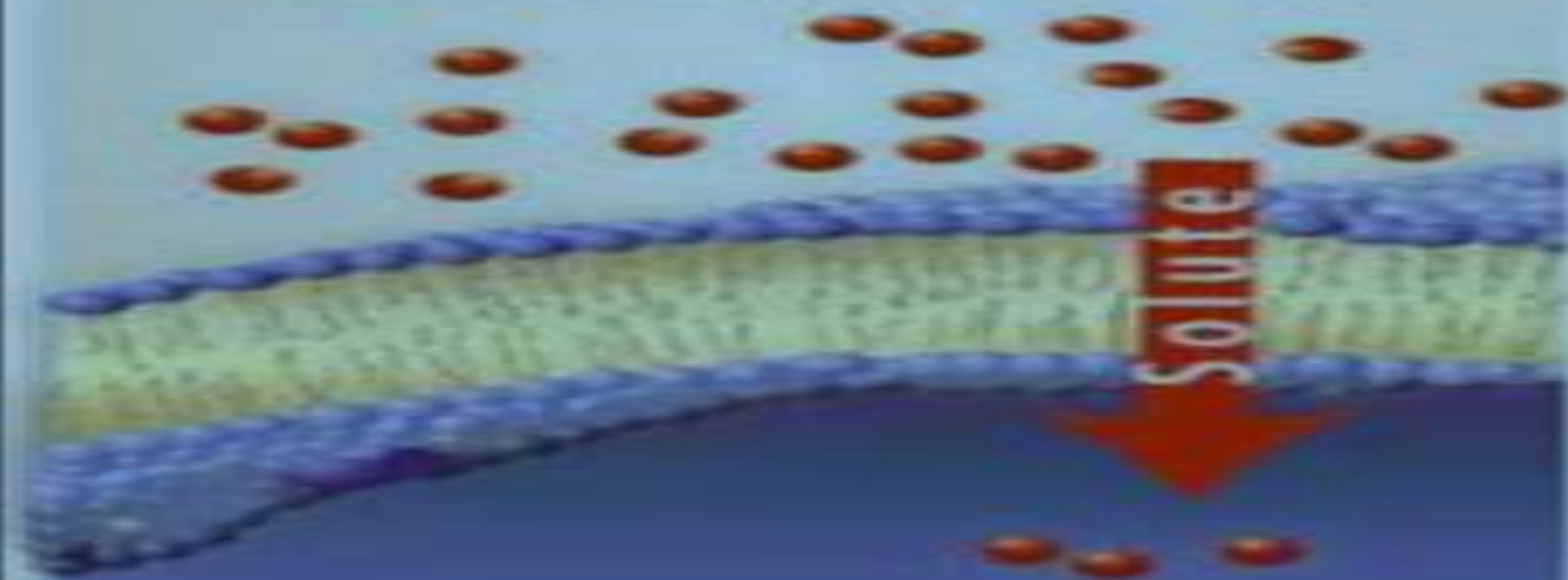
- Bulk flow
- Filtration



PASSIVE DIFFUSION

- Passive diffusion involves the movement of drug molecules down a concentration or electrochemical gradient without the expenditure of energy
- It is not saturable
- Small, lipid soluble, non-ionized molecules are easily absorbed by passive diffusion
- Passive diffusion is the most common and most important mode by which drugs cross biologic membranes

Higher solute
concentration



Lower solute
concentration

FILTRATION (BULK TRANSPORT, SOLVENT DRUG)

- Filtration is the bulk flow of solvent and solute through channels (pores) in the membrane
- Filtration is seen with small molecules (usually with a molecular weight less than 100) that can pass through pores
- Some substances of greater molecular weight, such as certain proteins, can be filtered through intercellular channels
- Concentration gradients affect the rate of filtration

Bulk flow and Filtration

- Bulk flow is an important way of a drug to move out of blood vessel
- Filtration is an important way to excrete drug

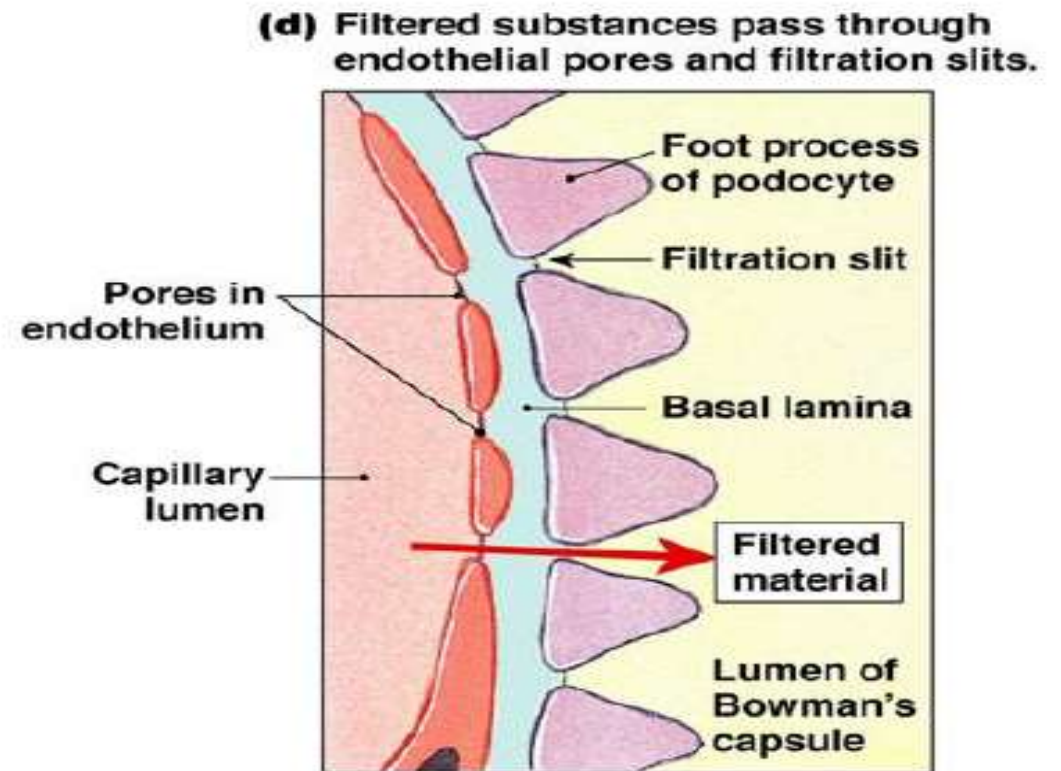
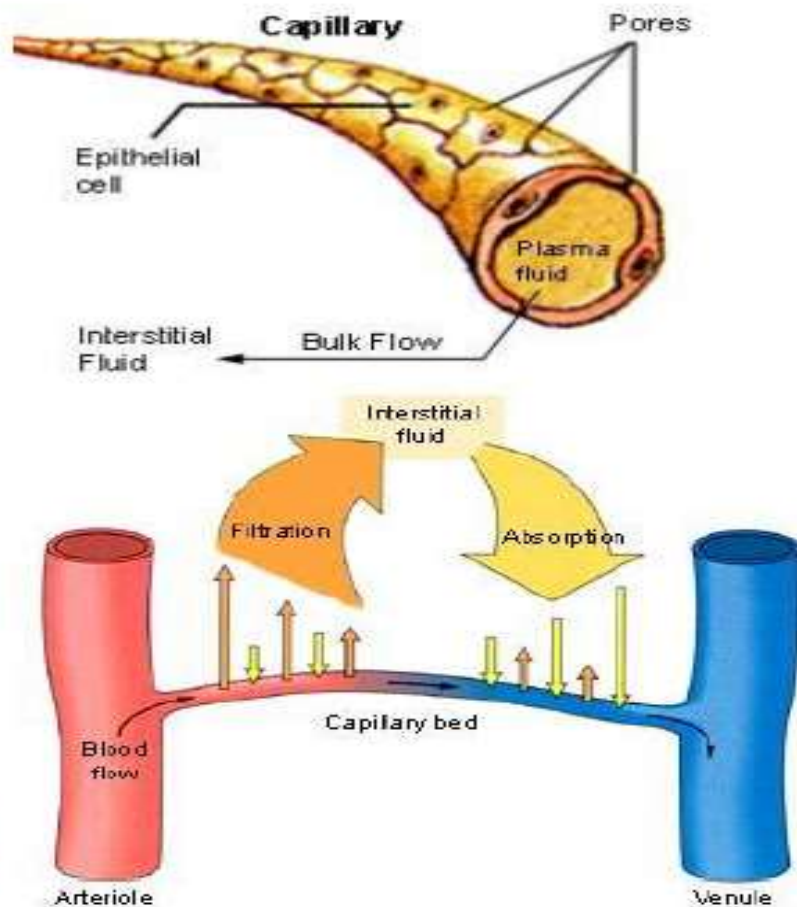


Fig. 19-4d

FACILITATED DIFFUSION

Facilitated diffusion involves the transport of a molecule along its concentration gradient through the facilitation of a carrier protein

Drugs and compounds that are too large and/or too polar to diffuse across a lipid membrane are transported by facilitated diffusion

Sugars and amino acids are transported in this way as are certain drugs (e.g. penicillin and quinine)

- Does not require energy
- The transporter protein may become saturated, compounds may compete for binding on the carrier protein and some compounds may inhibit it

ACTIVE TRANSPORT

Active transport involves the active movement of a drug or ion across a membrane and can move drugs against a concentration gradient

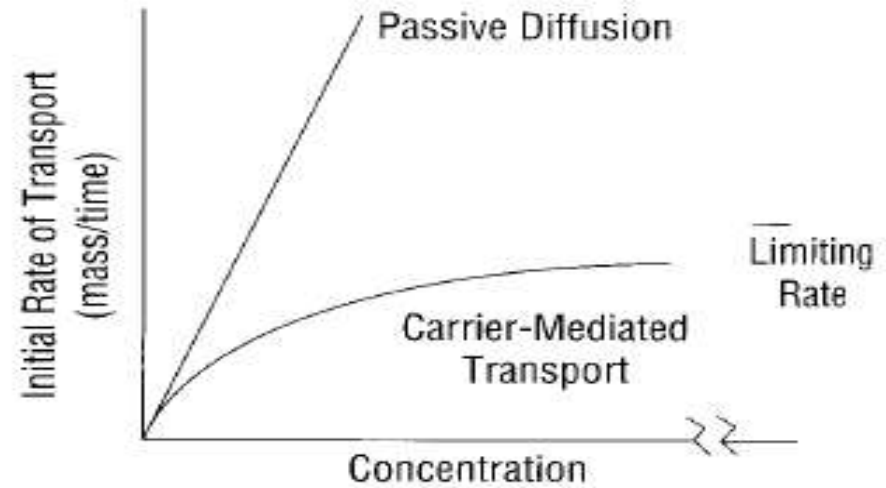
It is usually the mode of transport for drugs that resemble actively transported endogenous substances such as sugars, amino acids and nucleosides

- Involves the use of carrier proteins
- It occurs in only one direction
- Requires energy and is saturable
- Competition for binding and inhibition can occur

Carrier-mediated transport

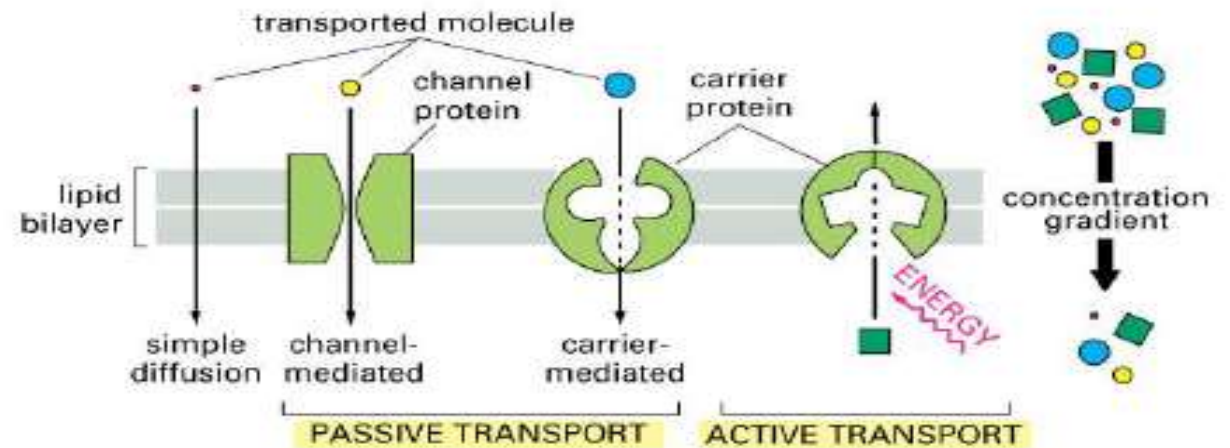
- **Facilitated diffusion**

- Carrier needed
 - Can be saturated
- No energy required
- Move along conc. gradient



- **Active transport**

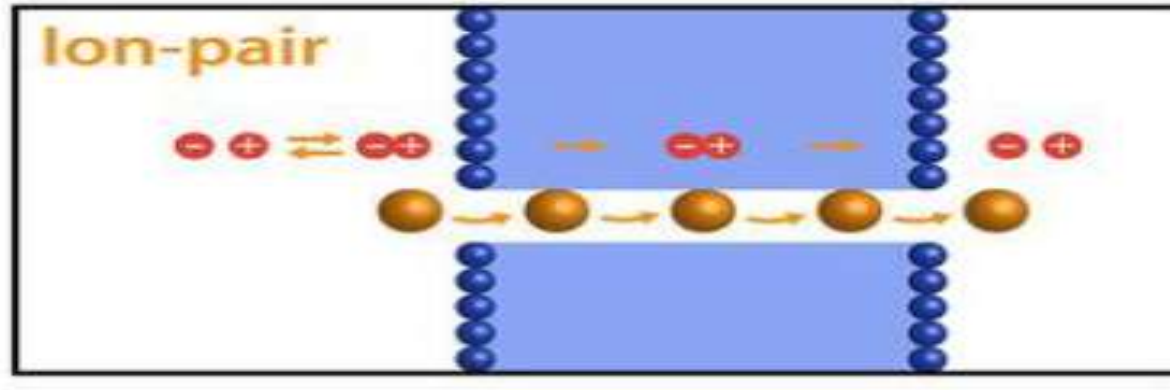
- Carrier needed
 - Can be saturated
- Energy required
- Move against conc. gradient



ENDOCYTOSIS

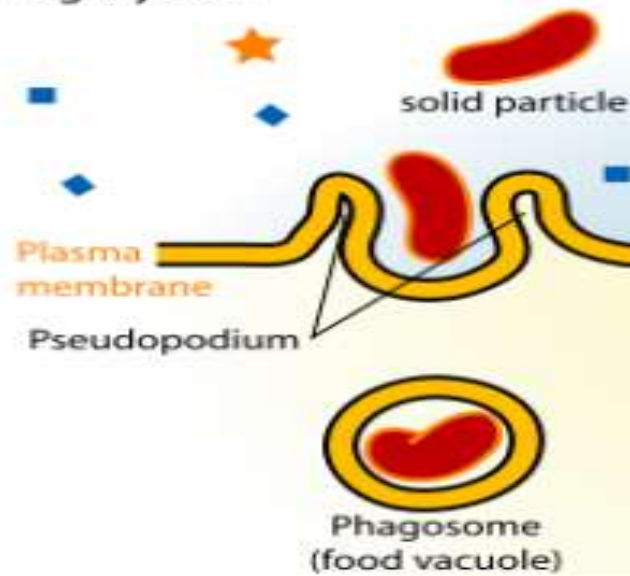
- Endocytosis is the process where small droplets are engulfed by the cell. Some proteins are taken up by this process.
- The reverse process, exocytosis, is responsible for secretion of many substances from cells, e.g. neurotransmitters stored in nerve endings.

Ion-pair transport and endocytosis

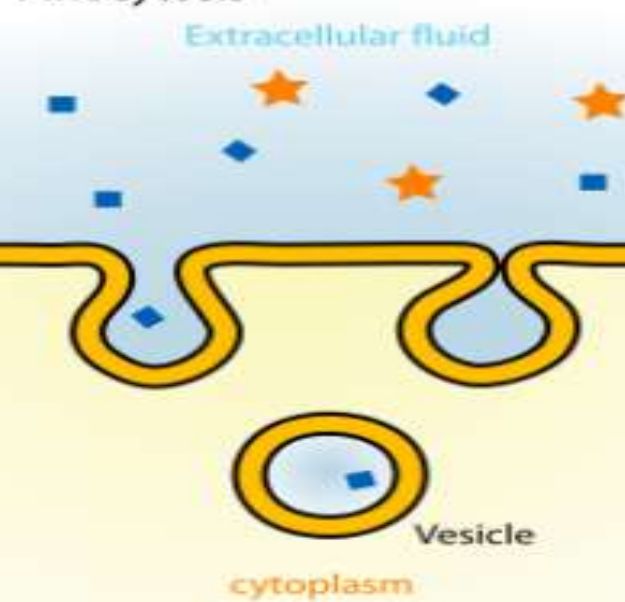


Endocytosis

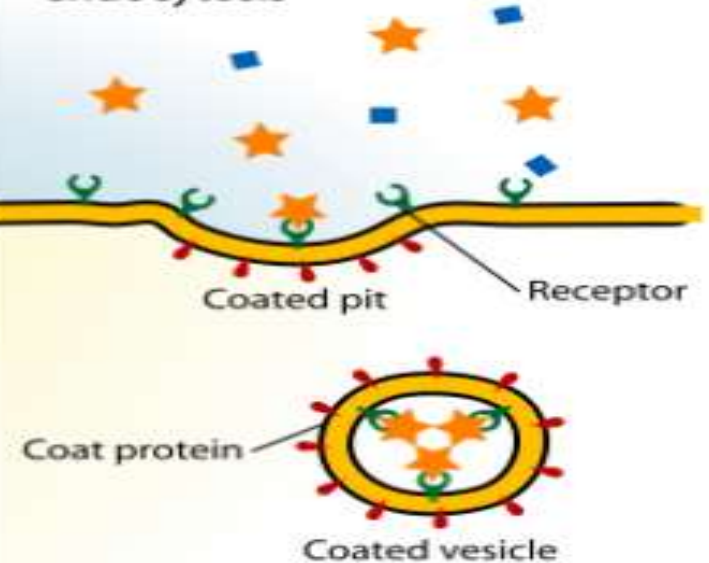
Phagocytosis



Pinocytosis

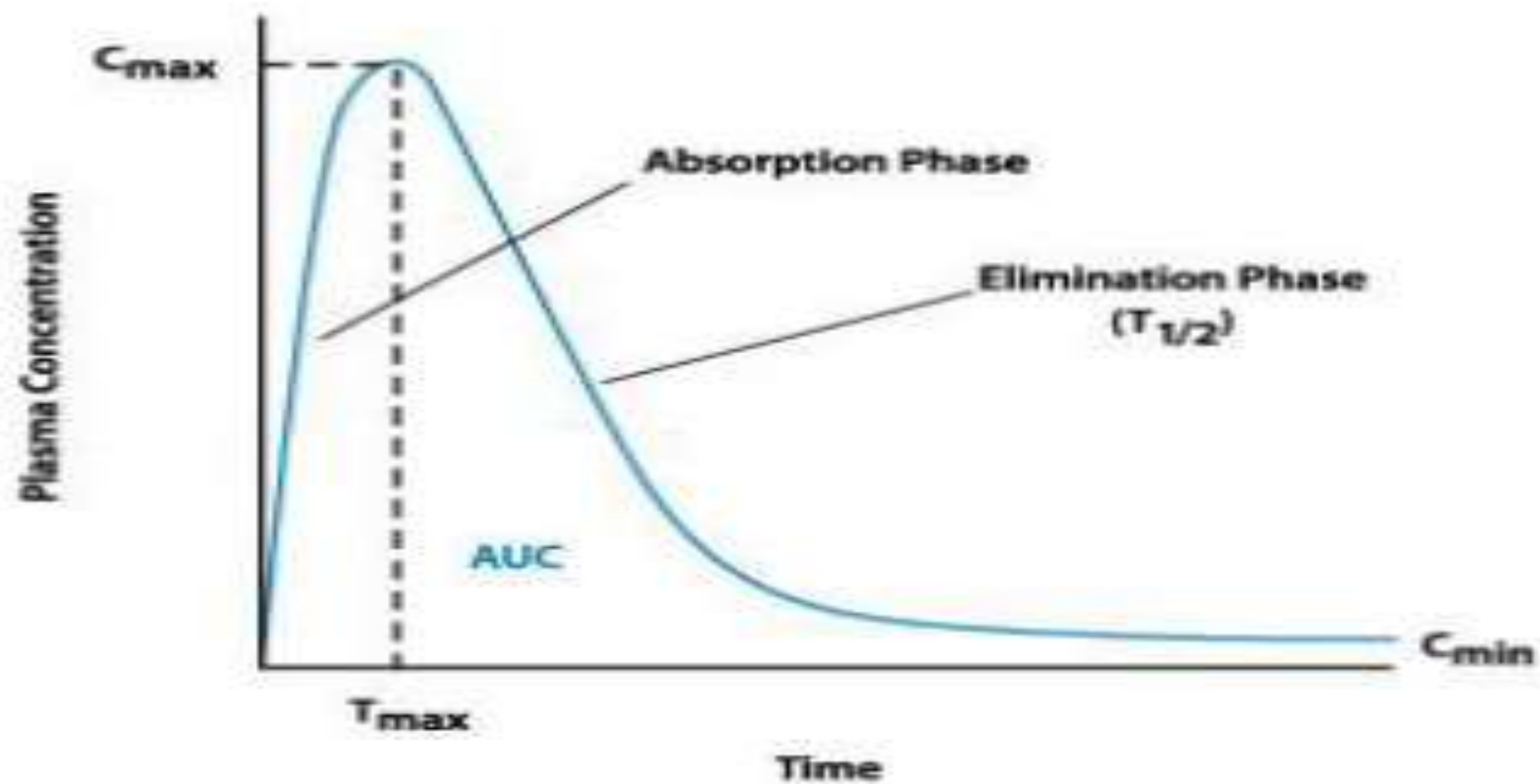


Receptor-mediated endocytosis



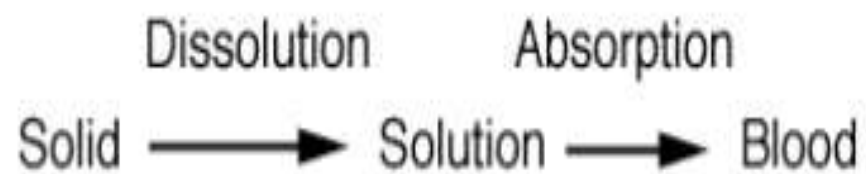
DRUG ABSORPTION

- Absorption is defined as the passage of the drug from the site of drug administration into circulation
- For a drug to reach its site of action, it must pass through various membranes depending on the route of administration
- Absorption occurs by one of the processes of membrane transport
- Except for the intravenous route, absorption is important for all other routes of administration
- Several factors influence the rate and extent of absorption of a drug

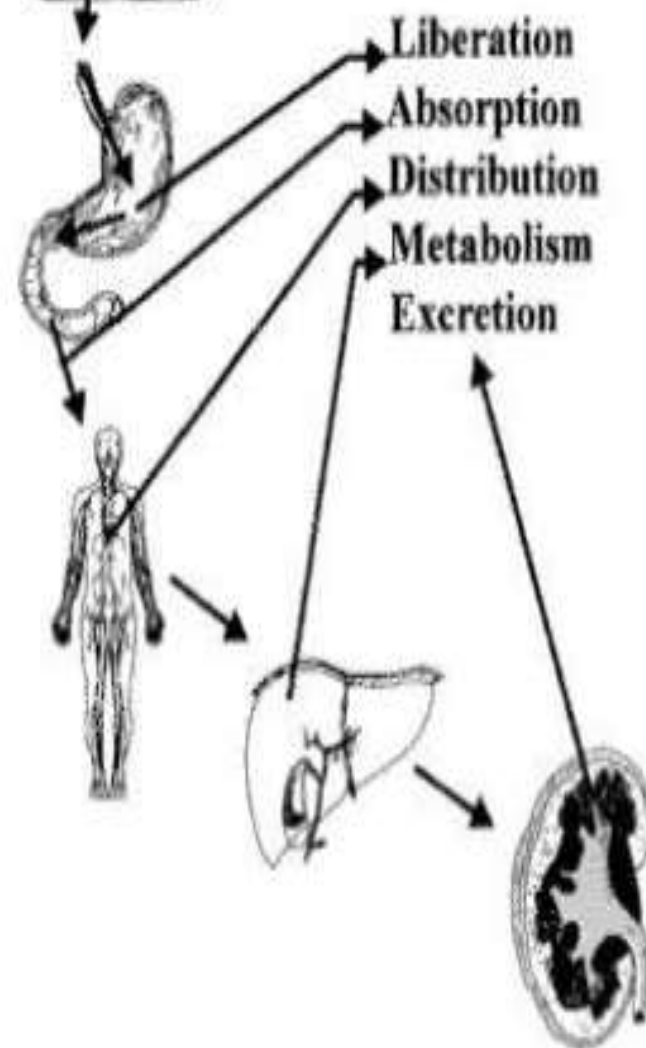


FACTORS THAT INFLUENCE THE RATE AND EXTENT OF DRUG ABSORPTION

- Disintegration and dissolution
- Formulation
- Particle size
- Lipid solubility
- pH and ionization
- Area and vascularity of the absorbing surface
- Gastrointestinal (GIT) motility
- Presence of food
- Metabolism
- Diseases of the gut



LADME



DISINTEGRATION AND DISSOLUTION

- Drugs taken orally should break up into individual particles (disintegrate) to be absorbed
- For drugs given subcutaneously or intramuscularly, the drug molecules have to dissolve in the tissue fluids
- Liquids are absorbed faster than solids
- Delay in disintegration and dissolution results in delayed absorption

FORMULATION

- Pharmaceutical preparations are formulated to produce desired absorption
- Inert substances used with drugs as diluents like starch and lactose may sometimes interfere with absorption

PARTICLE SIZE

- Small particle size is important for better absorption of drugs
- Drugs that are intended for action within the gut lumen have large particle size to reduce systemic absorption

LIPID SOLUBILITY AND IONIZATION

- Lipid soluble drugs are absorbed faster and better by dissolving in the phospholipids of the cell membrane
- Ionized drugs are poorly absorbed
- Unionized drugs are lipid soluble and are well absorbed
- Most drugs are weak bases and weak acids, and exist in both ionized and non-ionized forms. The degree of ionization depends on the pH.
- Strong acids and bases are highly ionized and therefore poorly absorbed

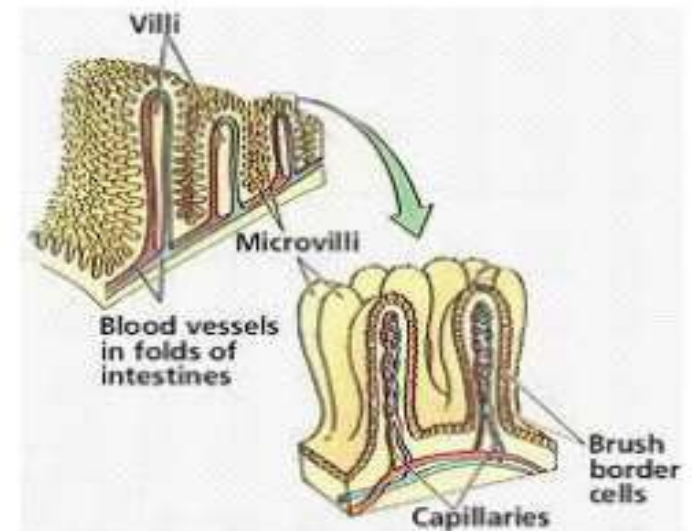
AREA AND VASCULARITY OF THE ABSORBING SURFACE

The larger the area of the absorbing surface and the more the vascularity, the better the absorption. Thus most orally administered drugs are absorbed from the small intestine.

- **Area of absorbing surface**

<u>Drug</u>	<i>% absorbed in 1 hr from stomach</i>	<i>% absorbed in 10 min from small intestine</i>
phenobarbital	17	52
pentobarbital	24	55
promethazine	0	38
ehtanol	38	64

Data from: Magnussen MP. *Acta Pharmacol Toxicol* 26:130, 1968.



GIT MOTILITY, PRESENCE OF FOOD AND GIT DISEASES

- If gastric emptying time is faster, the passage of the drug to the intestines is quicker and hence absorption is faster
- When intestinal motility is highly increased (e.g. in diarrhea), drug absorption is reduced
- Presence of food delays gastric emptying, dilutes the drug and delays absorption
- Drugs may form complexes with food constituents and such complexes are poorly absorbed
- Diseases of the GIT like malabsorption and achlorhydria resulting reduced absorption

METABOLISM IN THE GUT

- Some drugs may be degraded in the GIT
- Drugs that are metabolized in the gut should be given in higher doses or by alternative routes

LEARNING OUTCOMES

Students must be able to comprehend:

Processes involved in drug passage across biological membranes and in drug absorption

QUIZ

1. The main mechanism of most drugs absorption in GI tract is:
 - a) Active transport (carrier-mediated diffusion)
 - b) Filtration (aqueous diffusion)
 - c) Endocytosis and exocytosis
 - d) Passive diffusion (lipid diffusion)
2. What kind of substances can't permeate membranes by passive diffusion?
 - a) Lipid-soluble
 - b) Non-ionized substances
 - c) Hydrophobic substances
 - d) Hydrophilic substances
3. A hydrophilic medicinal agent has the following property:
 - a) Low ability to penetrate through the cell membrane lipids
 - b) Penetrate through membranes by means of endocytosis
 - c) Easy permeation through the blood-brain barrier
 - d) High reabsorption in renal tubules
4. What is implied by «active transport»?
 - a) Transport of drugs through a membrane by means of diffusion
 - b) Transport without energy consumption
 - c) Engulf of drug by a cell membrane with a new vesicle formation
 - d) Transport against concentration gradient

END

Thanks for listening